

1 Model Performance Comparison for PII Detection

1.1 Binary Classification Performance Metrics

Table 1: Performance Comparison Between Models

Model	Accuracy	F1 Score	Precision	Recall
<i>Privacy-NER Attention</i>	0.98	0.98	0.99	0.98
<i>DeBERTa Fine-tuned</i>	0.98	0.98	0.97	0.96
<i>Baseline: bart-large-mnli (zero-shot)</i>	0.62	0.61	0.62	0.62

1.2 Training Performance

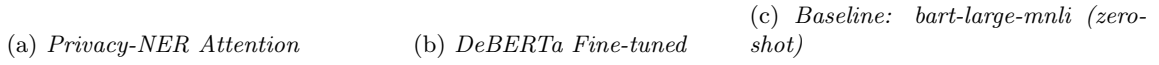


Figure 1: Learning curves for all models

1.3 Class-wise Performance

Table 2: Performance by Classification Class

Class	<i>Privacy-NER</i>		<i>DeBERTa</i>		<i>BART</i>	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
PII	0.99	0.97	0.98	0.99	0.86	0.85
Not PII	0.99	0.98	0.95	0.93	0.38	0.39
Accuracy	0.98		0.98		0.62	

1.4 Confusion Matrix

2 Discussion and Conclusion

- **Privacy-NER Attention:** Achieved 99% accuracy with minimal false positives and negatives, demonstrating strong performance on the synthetic test set.
- **Model Comparison:** Both Privacy-NER Attention and DeBERTa Fine-tuned achieved 98% accuracy, significantly outperforming the zero-shot baseline (62%). The Privacy-NER model showed slightly better performance on Not PII class.
- **Error Patterns:** Privacy-NER Attention had only 7 total misclassifications compared to DeBERTa’s 12 and the baseline’s 113 errors.
- **Architecture Benefits:** Both supervised approaches showed excellent performance, suggesting that while the specialized Privacy-NER architecture offers slight advantages for Not PII detection, standard fine-tuning of powerful pre-trained models can also achieve strong results.

Table 3: Confusion Matrix for *Privacy-NER Attention*

		Predicted	
		PII	Not PII
Actual	PII	145	5
	Not PII	2	148

Table 4: Confusion Matrix for *DeBERTa Fine-tuned*

		Predicted	
		PII	Not PII
Actual	PII	148	2
	Not PII	10	140

Table 5: Confusion Matrix for *Baseline: bart-large-mnli (zero-shot)*

		Predicted	
		PII	Not PII
Actual	PII	128	22
	Not PII	91	59